From routine cine CMR to regional myocardial dysfunction:

Fidelity, validation and interpretation of motion and strain estimates

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Abstract

Regional myocardial dysfunction can be clinically important while global ventricular measures remain preserved. Routine cine cardiovascular magnetic resonance (CMR) is widely available and supports retrospective motion and strain estimation, and what it records is changing image intensity and anatomy without labels for myocardial material points. Every estimator therefore adds assumptions to recover correspondence, and the question for a regional claim is which attributes of an abnormality, its magnitude, location, spatial extent and timing, are recovered within a tolerance appropriate to the intended use.

This critical narrative review compares feature-tracking, deformable-registration, biomechanical and learning-based estimators by their inputs, representations, priors and supervision, and groups the validation literature by the claim each target can support: numerical phantoms, tagging, displacement encoding with stimulated echoes (DENSE), scan–rescan studies, late gadolinium enhancement and clinical outcomes. Current evidence most directly supports cine-derived circumferential strain at whole-slice, global and segmental scales under matched definitions, including improved segmental agreement with DENSE after DENSE-supervised learning. Agreement is weaker for regional than global quantities and weakest for radial strain. The one independent test against a prescribed focal scar preserved slice-averaged circumferential strain better than radial strain without quantifying recovery of the scar region, and direct tests of focal boundary, extent and peak timing remain sparse.

Apparent contradictions resolve when representation capacity, estimation error and input sufficiency are separated and when estimands are matched. A defensible validation programme pre-specifies the measurement contract, combines known-motion tests with registered material-sensitive in-vivo comparisons, and ties each regional claim to the method, acquisition, population, scale and reference for which it was demonstrated.

Keywords: cardiovascular magnetic resonance; cine CMR; myocardial motion; myocardial strain; regional dysfunction; feature tracking; image registration; validation.

Plain English summary

The heart can contain a small area that moves abnormally even when its overall pumping measure looks normal. Standard cine CMR provides moving pictures of the heart and is available in routine care. Computer methods can use these pictures to estimate motion and strain, a measure of deformation. An

ordinary cine image, however, does not label each piece of heart muscle, so different methods can produce different internal motion while matching the same heart borders.

We reviewed how cine-based methods work and how they have been tested. Some studies show useful agreement with specialised CMR methods that are more sensitive to tissue motion. Other studies show important differences between software products, strain directions and heart regions. A good-looking contour, a repeatable number, an association with scar and a prediction of outcome are all valuable findings, and each of them answers a different question.

Regional measurements should therefore be judged for the exact property they are meant to report: the size, position, extent or timing of an abnormality. Tests with known motion should be combined with carefully registered comparisons to specialised scans in people. The method, scan settings, strain definition and acceptable error should be stated in advance. This approach can identify where routine cine CMR gives reliable regional information and where the result should remain qualified.

1 Why regional function is a distinct measurement problem

Left-ventricular ejection fraction compresses a complex three-dimensional deformation into a chamber-level ratio. That compression has real value: ventricular volumes and ejection fraction are reproducible CMR endpoints, widely understood and embedded in clinical reporting standards [1, 2]. The same compression also hides information. Well-functioning regions can compensate for a smaller abnormal region, opposing changes can cancel in an average, and the timing of regional contraction can change without a large fall in ejection fraction. Regional motion and strain are attractive precisely in the clinical settings where such hidden changes matter: focal infarction, heterogeneous cardiomyopathy, dyssynchrony, early dysfunction and treatment response.

Clinical studies support the relevance of measuring deformation in those settings. Feature-tracking strain has been associated with functional recovery after myocardial infarction, adverse remodelling and cardiovascular outcomes [3–5], and DENSE circumferential strain has shown prognostic association after acute infarction [6]. These associations justify the measurement. They leave open how accurately a regional abnormality has been located or quantified, because a prognostic marker can remain useful despite measurement bias, and a technically accurate motion field may still fail to improve a clinical decision. Analytical validity, biological association and clinical utility are linked questions with separate answers [7, 8].

Earlier reviews have covered the first of these questions at the level of strain physics, modality differences, clinical practice and sources of discrepancy [9, 10], and more recent reviews cover deep learning for wall-motion analysis across modalities and strain from CMR and computed tomography [11, 12]. The present review takes a narrower purpose within that literature. It asks how faithfully routine cine CMR can recover *regional or focal* myocardial dysfunction, and how the choice of validation target changes the claim that can be made. Answering that question requires comparing estimator families by their inputs, assumptions, estimands, populations and reference standards: how each family uses image information, representation, priors and supervision to infer motion and strain, and which spatial and temporal scales of measurement its validation evidence supports. Within that comparison, the focal case is the most demanding one, and it is used here as a test of the evidence base, without treating every method paper as if it had set out to solve it.

The practical question is:

> For a specified cine acquisition, estimator and measurement definition, which attributes of a regional abnormality (magnitude, location, spatial extent and timing) are recovered within a

tolerance that is appropriate for the intended use?

This formulation steers between two unhelpful extremes. At one extreme, routine cine would be treated as information-free; in reality myocardial borders, papillary muscles, trabeculation, intensity variation, neighbouring structures and temporal consistency all constrain motion. At the other extreme, a visually plausible deformation or an accurate contour would be treated as proof of correspondence; in reality neither uniquely identifies how material points moved within the myocardium. Between these extremes, the evidence supports bounded successes, and the review is organised to state those bounds.

1.1 Review approach and scope

This is a critical narrative review. Its question is what routine cine CMR can establish about regional myocardial motion and strain, so the literature was assembled to cover the estimator families, the validation targets and the public resources that bear on that question; it does not enumerate every cine strain study. The corpus was built in three steps. The first step began from a core set of papers, listed in the Supplementary Methods, and followed their reference lists and citing articles; guideline, dataset and repository records were added by hand because bibliographic databases index them incompletely; expansion stopped when a further round changed no method family, comparison axis or principal conclusion.

The second step was a structured search of PubMed, Europe PMC and OpenAlex, with a publication cutoff of 1 September 2026, built from six concepts: regional cine strain after infarction, known-motion or digital-phantom validation, perturbation of the prior or regularisation, conditional recoverability (whether a regional abnormality can be recovered under stated acquisition and estimator conditions), failure-aware or selective prediction, and regional localisation, spillover and attenuation. The purpose of that search was to find any study that combined a prescribed regional abnormality, known motion and a controlled change of the estimator, and to collect the studies that address part of that design. Records were screened on title and abstract against this purpose, and each retained record that supports a statement in the review was then read at the depth recorded in the evidence ledger. The third step, completed on 8 October 2026, re-verified the central primary studies, the dataset descriptions and the bibliographic metadata of every reference against PubMed/PMC, publisher pages, Crossref or DataCite records and official dataset pages, and read the reviews published from 2024 to 2026, which also formed part of the core set, so that this review could be positioned against them.

Three limits follow from this design. The structured search is bounded by its query strings, its three databases and a cap of 100 records per query, so the statements in Section 3.3 and Section 4.1 that no study of a given kind was identified hold within that boundary. Screening was performed in a single pass with software assistance for query execution, deduplication and screening suggestions (see Declarations); it was not repeated by a second reviewer, no record was excluded on full-text criteria, and the retention decisions are the author's responsibility. Reading depth depends on the role a source plays: the primary studies on which the principal conclusions rest were read in full; part of the method-lineage and reconstruction literature was read at abstract level, and an abstract-level reading supports only the study design and the results stated in the abstract. The evidence ledger states the depth for every cited source, and Supplementary Table S3 marks the entries read at abstract level. The query strings, hit counts and screening record are given in the Supplementary Methods.

The studies discussed in Sections 3 to 6 were compared along six axes: (1) image input and sampling; (2) population and abnormality; (3) estimated entity, such as contour motion or dense displacement; (4) strain definition and aggregation; (5) validation target; and (6) reported error or association. Because these axes, reference standards and endpoints were too heterogeneous to share a scale, results were not pooled.

Dice overlap, displacement error, strain bias, scan–rescan precision, scar discrimination and outcome association were kept as different quantities throughout.

2 From cine images to a regional measurement

2.1 What routine cine CMR observes

Balanced steady-state free-precession cine CMR samples a time series of image intensities. In a typical short-axis image, the blood–myocardium interfaces are conspicuous, whereas intramyocardial texture can be weak and changes with coil sensitivity, through-plane motion, partial volume, reconstruction and artefact. The image therefore supplies anatomical and intensity cues without a persistent identity label for every tissue element. Every estimator has to add something to those cues. Feature tracking exploits image features or propagated contours; registration exploits an image-similarity relation; biomechanical models constrain admissible deformation; and learning-based methods import associations learned from training data [13, 14]. These four categories overlap in practice: a neural network may predict registration parameters, a registration method may use a learned similarity measure, and a biomechanical penalty may be embedded in either.

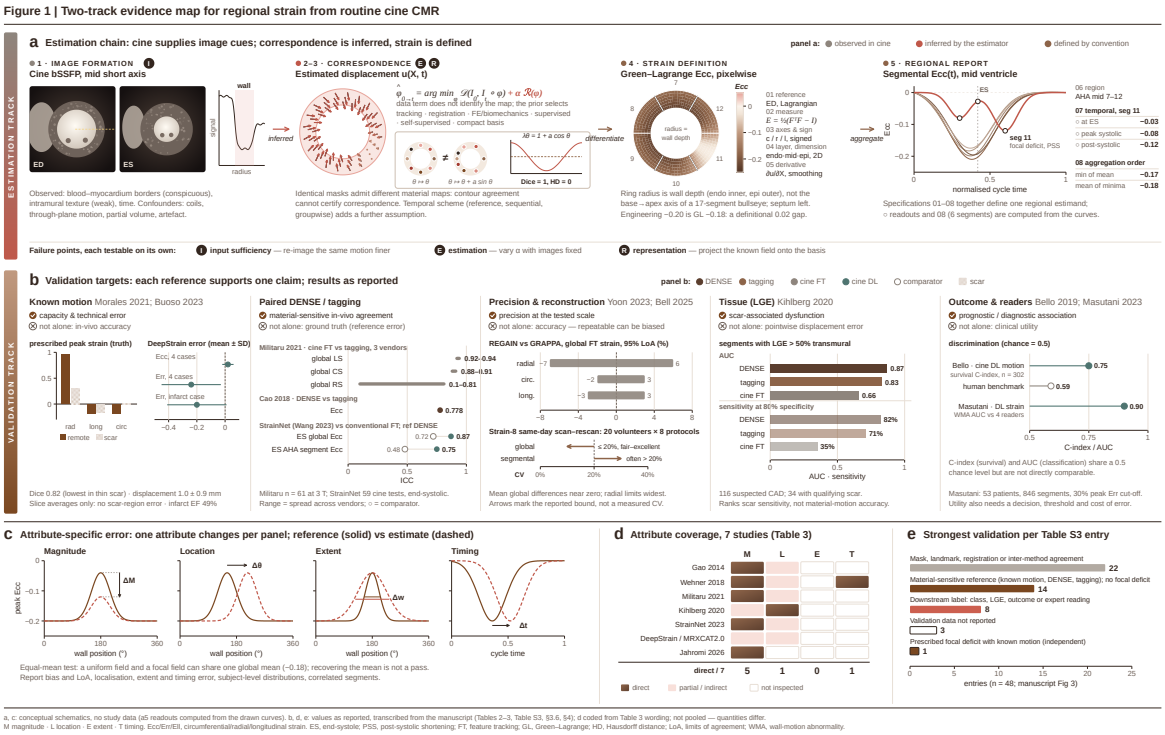


Figure 1. Two-track evidence map for regional strain from routine cine CMR. (a) Estimation track: cine supplies image cues (blood–myocardium borders, weak intramural texture and time), an estimator infers material correspondence from them, and strain is then fixed by convention through eight specifications (reference, measure, axes and sign, layer and dimension, derivative and smoothing, region, temporal readout and aggregation order). The circled letters mark three failure points, each testable on its own: input sufficiency (I), by re-imaging the same motion more finely; estimation (E), by varying the regularisation weight α with the images fixed; and representation (R), by projecting the known field onto the estimator's basis. (b) Validation track: each reference supports one claim and leaves the next open on its own (known motion, paired DENSE or tagging, precision and reconstruction, tissue characterisation, and outcome or reader labels), with results as reported in the cited studies. (c) Attribute-specific error: magnitude, location, extent and timing each change one attribute of a reference profile (solid) against an estimate (dashed); a uniform and a focal field can share one global mean. (d) Direct inspection of magnitude (M), location (L), extent (E) and timing (T) in seven paired or regional studies, coded from the wording of Table 3. (e) Strongest validation evidence per Supplementary Table S3 entry (n = 48, as in Figure 3). Panels a and c are conceptual schematics without study data, and the readouts in a are computed from the drawn curves; the values in b, d and e are transcribed from the tables and sections of this review and are kept unpooled because the quantities differ. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

The separation between observed cues and inferred correspondence in Figure 1a is a causal one, and it can be written down. Let a point in a reference configuration be $\boldsymbol{X}$, and let an estimated mapping be $\boldsymbol{\varphi}(\boldsymbol{X}, t)$. The displacement and deformation gradient are

$$\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{\varphi}(\boldsymbol{X}, t) - \boldsymbol{X}, \qquad \boldsymbol{F}(\boldsymbol{X}, t) = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}} = \boldsymbol{I} + \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}}. \tag{1}$$

One possible finite-strain measure is the Green–Lagrange tensor,

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^{\mathsf{T}}\boldsymbol{F} - \boldsymbol{I}). \tag{2}$$

These equations (Equation (1) and Equation (2)) state how a chosen mapping is converted into a chosen strain; they say nothing about whether $\boldsymbol{\varphi}$ is correct. The conversion itself has consequences for error. Differentiation can amplify local mapping error, while smoothing can reduce variance and alter peaks or

spatial gradients. The reported number further depends on coordinate axes, myocardial layer, reference frame, segment definition, spatial aggregation and whether "peak" means the minimum of a segment-average curve or the average of pointwise minima. A comparison that ignores these choices can attribute a definitional difference to a tracking error [15, 16].

2.2 The same contours can support different material mappings

Boundary agreement constrains deformation without determining it. Consider an annulus whose end-diastolic and end-systolic contours are fixed. Material points can retain their angular positions, or they can be reparameterised around the same final contour. Both mappings give Dice overlap of one and Hausdorff distance of zero for the masks, yet their circumferential stretch varies differently with position (Figure 2). The counterexample proves a limited point: equal masks or equal mean geometry are insufficient to identify intramyocardial correspondence. It leaves two other questions open, namely whether cine-derived regional strain is wrong in any particular case, and whether image texture and temporal information can select a useful mapping.

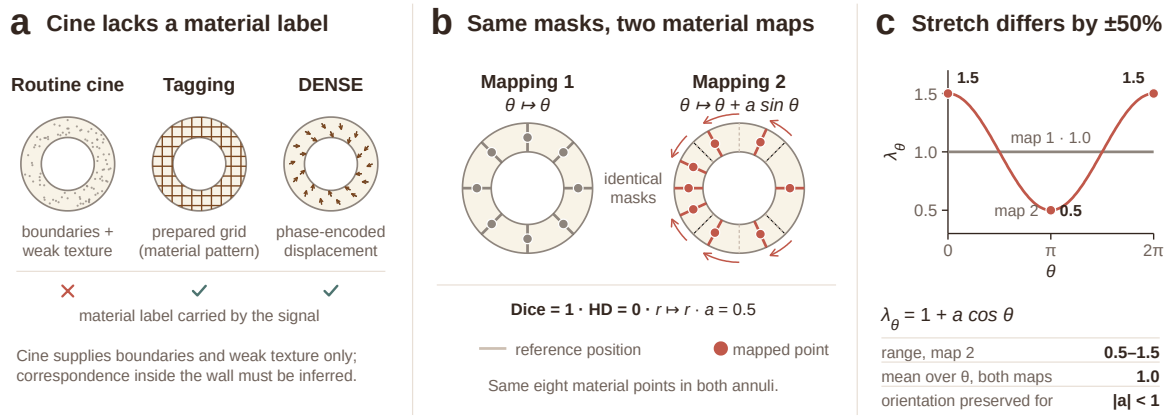


Figure 2. Identical contours do not determine the same material correspondence. (a) Schematic of the information each acquisition supplies: routine cine provides boundaries and weak texture, tagging a prepared pattern, DENSE phase-encoded displacement; only tagging and DENSE carry a material label in the signal. (b) Two material mappings between the same annular contours ($r \mapsto r$): angular identity, $\theta \mapsto \theta$, and $\theta \mapsto \theta + a \sin\theta$ with $a = 0.5$. Both give identical masks (Dice = 1, Hausdorff = 0). (c) Their circumferential stretch, $\lambda_\theta = 1$ and $\lambda_\theta = 1 + a\cos\theta$, differs; the latter ranges from 0.5 to 1.5 and preserves orientation for $|a| < 1$, and both maps have the same mean stretch over θ, 1.0. The figure is an analytic counterexample; no estimator error is measured, and the stretch ratio is a different quantity from Green–Lagrange strain. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

The consequence of this non-uniqueness grows as the claim becomes more local. Chamber volume can be estimated from boundaries alone. A global contour-length change is also strongly boundary-constrained. A segmental average additionally requires correspondence between sectors. Transmural or voxelwise strain requires, on top of that, intramyocardial correspondence and spatial derivatives. Along this sequence, a method can be accurate for volumes and useful for global strain while remaining insufficiently validated for focal strain gradients. Validation should therefore follow the finest claimed spatial and temporal scale; credibility earned at a coarser scale does not transfer downward.

2.3 A complete estimand is more than the word "strain"

Because each step of that sequence adds a definitional choice, a regional strain estimand should state at minimum:

- the reference configuration and mapping direction;
- the finite-strain measure (for example, linear, Green–Lagrange or logarithmic strain);
- the anatomical coordinate system and sign convention;
- the myocardial layer and dimensional model (2D slice, 2D+time or 3D+time);
- the differentiation, interpolation and spatial regularisation used;
- the region definition, registration to the AHA model and treatment of junctions;
- the temporal summary, including end-systolic versus peak strain and handling of post-systolic shortening; and
- the order of spatial and temporal aggregation.

When these items are missing, apparent disagreement may reflect different quantities. The problem is most prominent for radial strain, for which wall-thickening surrogates, tensor components and layer-specific measurements are not interchangeable. It also affects circumferential comparisons whenever one pipeline reports one-dimensional engineering strain and another reports a component of a two-dimensional Green–Lagrange tensor [16, 17].

3 Estimator families and the assumptions they contribute

All estimators resolve incomplete image information by adding structure. The useful comparison between them therefore concerns which assumptions each introduces, whether those assumptions are visible to the user, and whether they suit the abnormality being studied. Table 1 summarises these attribute-level distinctions. Section 3.1 then states the estimation problem that all the families share, and Section 3.2 to Section 3.6 follow the literature in approximate chronological order within each family. A year-ordered table of the individual methods, with the data, validation approach and pathology evaluation each source reports, is given in Supplementary Table S3.

Table 1. Method attributes that may coexist within a cine-CMR motion estimator. Rows are descriptive archetypes; a single method may combine several of them.

Archetype	Input	Output and estimand	Motion representation	Constraint/prior	Optimisation or training target	Endpoint directly validated in representative work
Feature and contour tracking	Cine pixels plus initial contours	Contour/mesh motion; product-defined strain	Boundary points, meshes or proprietary field	Feature similarity and trajectory continuity	Frame-to-frame match or contour propagation	Global/segmental peak strain, repeatability and tissue discrimination; product-specific
Deformable registration	Image pairs or cine sequence	Dense displacement; derived strain	Voxel field, spline, spectral or implicit field	Spatial/temporal regularity; optional topology	Image/feature similarity plus penalties	Image/shape alignment; selected phantom, tagging or DENSE comparisons
FE model	Images, masks or meshes	Nodal displacement and tensor strain	Finite-element or reduced-order basis	Compatibility, volume constraint, constitutive/fibre terms	Image correlation plus mechanical residual	Landmark/curve or phantom error in the specified model domain
Supervised learning	Cine or contours plus motion/strain labels	Learned displacement or strain	Dense field or parameter map	Architecture, augmentation and label domain	Label loss; optional image, shape or mechanics terms	Depends on label: StrainNet separates DENSE-contour EPE from cine-contour Ecc
Self-supervised learning	Cine without motion labels	Dense displacement/strain	Usually voxel or multi-scale field	Smoothness, cycle, topology or shape	Image, feature or mask agreement	Commonly registration and anatomy; material endpoints require separate testing

Archetype	Input	Output and estimand	Motion representation	Constraint/prior	Optimisation or training target	Endpoint directly validated in representative work
Compact basis	Any compatible image estimator	Coefficients decoded to motion/strain	Fourier, spline, temporal or neural basis	Band limit or basis dimension	Inherits estimator objective; may be learned	Capacity and final estimator performance are distinct endpoints

3.1 An underdetermined problem, and what the prior is doing

Let $I_t(\boldsymbol{x})$ denote image intensity at position $\boldsymbol{x}$ in phase t, and let $\boldsymbol{\varphi}_{0\to t}$ map a material point in the reference configuration to its position at t. Almost every estimator reviewed below can be written as, or interpreted through, the objective

$$\widehat{\boldsymbol{\varphi}}_{0\to t} = \arg\min_{\boldsymbol{\varphi}_{0\to t}} \mathcal{D}(I_0,\, I_t \circ \boldsymbol{\varphi}_{0\to t}) + \alpha\, \mathcal{R}(\boldsymbol{\varphi}_{0\to t}), \tag{3}$$

in which $\mathcal{D}$ rewards image agreement, $\mathcal{R}$ encodes prior structure and α sets their balance. The map is written here as a forward map on the reference domain. Frameworks that optimise a backward sampling map use a different convention, and the two conventions must not be interchanged when transformations are composed or differentiated for strain. The same objective covers the other families: groupwise and sequence models optimise an analogous expression over the whole cycle at once, and learning-based methods minimise its expectation over a training set in place of solving it per case.

The difficulty is that $\mathcal{D}$ alone does not identify the map. Where intramyocardial texture is weak, many fields achieve nearly the same data term while implying different displacement gradients, and the prior selects among them; Figure 2 is the analytic instance of this non-uniqueness. The prior's role can be read in two ways, and the review keeps them separate throughout. The first concerns what a representation can *express*. Strain is a spatial derivative of displacement, so a displacement field that looks smooth can still carry a rapidly varying strain. Restricting the field to a band of spatial frequencies restricts the derivatives it can represent, and the size of that restriction depends jointly on the bandwidth, the spatial scale of the abnormality and the measurement definition. The second concerns the estimation *process*. Even with a representation rich enough to express a focal deficit, the optimiser or network may fail to recover it if the image evidence is weak or the penalty $\alpha\mathcal{R}$ is strong. A third factor, the information actually present in the input images, is separate from both.

These three factors come apart in practice, and the way they come apart is the basis of an attribution experiment. Suppose a focal deficit is under-recovered. If the same estimator recovers it when α is reduced, the limitation was the estimation process. If it is still under-recovered after the known deficit has been projected directly onto the estimator's function space (a cheap test that uses the known field in place of the images), the limitation lies in the representation. If neither holds, and the deficit becomes recoverable when the same motion is re-imaged at finer resolution or with added long-axis views, the limitation lay in the acquired input. That last attribution is conditional, because a finer acquisition also changes discretisation, optimisation conditioning and, for a learned estimator, the training distribution, so the estimator, representation and training regime must be held fixed for it to hold. In every case, attributing an observed regional error requires an experiment that holds the other two factors fixed. Section 5.3 returns to this separation when reconciling discordant results, and the known-motion resources in Section 6.5 are the setting in which the experiment can be run.

Temporal strategy adds a further assumption to the objective. Reference-to-frame tracking keeps one reference configuration and must therefore estimate large displacements from it; sequential tracking estimates smaller inter-frame steps and accumulates composition error and drift; groupwise methods estimate all phases jointly under whatever temporal constraint they impose [18, 19]. None of these

strategies guarantees correct intermediate phases, because a periodic deformation can close perfectly at the end of the cycle and still be wrong in between. Cycle closure is therefore a useful constraint and a useful quality-control signal, and no more than that.

3.2 Feature tracking and contour propagation

Feature-tracking software is attractive because it operates on routine cine and fits existing clinical workflows. Initial contours are propagated through the cardiac cycle, after which strain may be calculated from endocardial length, a myocardial mesh or a vendor-specific deformation field; interior motion is to a large degree interpolated from boundary behaviour [13]. Because each of these steps can be implemented differently, the phrase "feature tracking" denotes a family of measurements. Commercial implementations have differed in contour placement, transmural treatment, radial-strain calculation, temporal interpolation and user correction. As a consequence, values may fail to be interchangeable across products or software versions even when repeatability within a product is acceptable [20, 21].

The family's clinical validation began before deep learning. Maret and colleagues related cine-derived functional measurements to scar in 30 patients after ST-elevation infarction [22]; Hor and colleagues compared mid-ventricular whole-slice peak circumferential strain with HARP in 191 patients with Duchenne muscular dystrophy and 42 controls [23]; and Augustine and colleagues studied 145 healthy volunteers with a tagging comparison in 20, finding closer global circumferential than longitudinal or radial agreement [24]. These are abstract-level readings, and the three studies validated three different regional quantities. Deep learning entered the family through its front end. Vigneault and colleagues segmented the sequence with a fully convolutional network and fitted quadratic basis splines across all frames in 42 patients with hypertrophic cardiomyopathy and 21 controls [25], and Puyol-Antón and colleagues combined convolutional segmentation with subsequent feature tracking and strain calculation [26]. In both designs the learned component is the segmentation, and the motion model remains a conventional tracker.

The strongest interpretation of a contour-tracking result depends on what was actually tracked. A reliable change in endocardial length can support a global or segment-level contour-derived strain under that definition; it cannot by itself support transmural material strain. The converse caution also applies: a method that uses internal image information should be described by its documented algorithm, and a product-category label such as "border tracking" should not replace that description. Method descriptions and validation endpoints, then, should govern interpretation. One evaluation caveat applies to this whole section. Segmentation can initialise the search region, define the cardiac coordinate system and, in several learning-based designs, supply an anatomical loss during training. When the same reference contours both shape the fitted transformation and provide the overlap score that assesses it, the evaluation is not independent of the fitting.

3.3 Conventional and learned registration, and the field representation

Intensity-based deformable registration estimates a dense field by optimising Equation (3) directly. The free-form deformation model of Rueckert and colleagues, an affine transform plus a B-spline control lattice with a smoothness penalty, remains the reference implementation; it was introduced for breast MRI [27]. Groupwise variants remove the dependence on an arbitrarily chosen reference frame. Metz and colleagues fitted an nD+t B-spline model to the whole dynamic sequence with spatial and temporal smoothness and an optional cyclic constraint, reporting sub-voxel accuracy on synthetic and thoracic CT data [18]. More recently, Chen and colleagues replaced the groupwise dissimilarity term with a locally low-rank metric and evaluated it on a 20-case simulation, 100 public cine cases and 16 in-house tagging patients. They reported reduced late-diastolic drift and better correlation with tagging for radial

strain in several segments than optical flow, pairwise registration, global low-rank groupwise registration and *n*D+*t* B-splines [19]. Optical flow itself offers another route to local correspondence: Wang and colleagues combined intensity and shape-feature invariance in a gradient-based method for cine and tagged MR, evaluated on realistic simulations and preliminary clinical data [28]. Two further designs supply correspondence through geometry or mechanics in place of image similarity: a Laplace-based mapping between successive contours [29] and variational assimilation of cine observations into a contraction model [30]. Each was tested on synthetic data with in-vivo feasibility, and in both the inferred trajectory depends on the modelling assumption. Anatomy-aware, pyramidal and online-adaptive methods change the problem along the same lines, by altering the evidence available to the optimiser and the class of admissible fields [31–33].

The learning-based literature began by amortising Equation (3). VoxelMorph reformulated registration as a function, parameterised by a convolutional network, that maps an image pair to a deformation field; the unsupervised training objective is the same image-matching and smoothness objective that a classical method optimises per pair, and the gain is that inference becomes a single forward pass [34]. TransMorph replaced the convolutional encoder with a hybrid Transformer–convolutional architecture and added diffeomorphic and Bayesian variants, validated on brain MRI and phantom-to-CT registration [35]. In this family the prior is still smoothness or diffeomorphism, now partly learned, and the reported evidence is label overlap and deformation-field regularity.

A distinct strand constrains the representation in place of the penalty. B-splines, finite-element bases, Fourier coefficients, implicit neural representations and dense voxel grids have different capacity and conditioning. Fourier-Net predicts a band-limited representation of the displacement in a low-dimensional frequency domain and recovers the full-resolution field by zero-padding and an inverse discrete Fourier transform [36]; Fourier-Net+ takes band-limited representations of the images themselves as input, with cascaded and diffeomorphic variants [37]. Band-limiting is an explicit, tunable statement about which deformations the model may output, which makes this family unusually well suited to the attribution experiment of Section 3.1. The bandwidth is, however, a global property of the field, so what it implies for a focal regional strain has to be measured. Both networks are general-purpose registration methods. Fourier-Net+'s published evaluation covers five datasets, including one 3D cardiac MRI dataset on which its diffeomorphic variant outperformed TransMorph and other baselines in Dice, Hausdorff distance, end-diastolic volume and ejection fraction; these are registration, geometric and functional measures, and they do not by themselves establish regional material-strain fidelity. Within the searches recorded here, this review did not identify an evaluation of a band-limited cine estimator against a regional motion or strain reference; that statement describes what the searches returned, and it does not assert that no such study exists.

3.4 Cardiac-specific designs: joint tasks, multiple views, meshes and time

A second group builds cardiac structure into the estimator. Qin and colleagues learned motion estimation and segmentation jointly, sharing a multi-scale encoder between an unsupervised Siamese recurrent spatial-transformer branch and a segmentation branch, in 220 subjects [38]. Zheng and colleagues learned a pixel-wise "apparent flow" between two cine phases semi-supervised from a large unsegmented and a small expert-segmented set, then derived per-segment radius and thickness time series whose nine features classified ACDC pathology with 95% and 94% accuracy on the training and test sets [39]. In that study the motion representation is validated through its usefulness for a downstream label, a pattern that recurs in Section 3.6. Motion Pyramid Networks introduced multi-scale fused estimation with a cyclic teacher–student training procedure [32], and FOAL added online adaptation of a learned estimator

to a new sequence at test time [33]. A 2025 semi-supervised design uses all frames of the sequence with an end-diastolic distance map weighting the loss, reporting an end-point error of 1.02 mm against 1.19 mm for a retrained RAFT baseline in an in-house set of 271 patients; its strain endpoint is global peak correlation, without a material reference [40].

Multi-view and mesh-based designs target through-plane motion. MulViMotion fuses short-axis and long-axis cine in a hybrid 2D/3D network with a shape-regularisation module providing weak supervision, evaluated on 580 UK Biobank subjects [41]. DeepMesh represents the heart as a 3D mesh of endocardial and epicardial surfaces, propagates a template mesh into subject space and estimates vertex-wise displacement through a differentiable mesh-to-image rasteriser, so that vertex correspondence is maintained across frames and subjects [42]. CardioMorphNet takes the shape idea further: a recurrent variational autoencoder with separate posteriors for segmentation and motion, trained by recursively registering segmentation maps without an intensity-based similarity loss, validated on UK Biobank and M&Ms by comparing warped masks with reference masks and yielding voxel-wise uncertainty maps [43]. Predictive uncertainty and accurate warped masks are valuable outputs, with two limits: the uncertainty is not automatically a calibrated regional error, and a Bayesian output is not automatically an abstention policy. STCMT-Net represents cardiac motion through keypoints extracted without supervision and reconstructs the dense field as a weighted combination of basis vectors from matched keypoint pairs, evaluated on 4D CT and cine MRI with a strain analysis reported to separate patients under different conditions [44].

Temporal structure has been addressed in parallel. Bidirectional recurrent registration predicts reference-to-frame motion with sequence context [45]; SegMorph couples motion and segmentation in a recurrent variational representation [46]; and an implicit-guided bidirectional residual framework reports improved anatomical registration in its retrieved abstract, with regional strain left as a separate question [47]. Further variants include a low-rank groupwise deformation model for cine tracking in a preprint that is a master's thesis and has not been peer reviewed [48], and a Mamba-based sequential model presented in a MICCAI workshop [49]. A continuous-time latent neural ordinary differential equation adds bidirectional forward–backward evolution and a Lagrangian consistency term, evaluated on ACDC and M&Ms [50]. One parallel development belongs here because of what it leaves out. CMR-specific foundation models now exist, one pre-trained on 15 million cine images from 74,916 studies [51] and another trained by self-distillation on 36 million images from 27,524 subjects [52], each fine-tuned for segmentation, landmark localisation, diagnosis and related tasks. Dense motion estimation is absent from the tasks either reports, so the pre-trained representations they offer have not yet been tested on the problem this review is about.

Read together, this group has made real progress on three-dimensional motion from two-dimensional acquisitions and on temporal consistency across the cycle. Its reported validation needs stating plainly. In the cardiac-specific papers reviewed here, the dominant evidence is agreement between a warped segmentation and a reference segmentation, together with surface distances and deformation-field regularity, with landmark error where a landmarked dataset was available. That evidence is appropriate for the claims those papers make, and it is also the evidence that Figure 2 shows cannot by itself establish material correspondence.

3.5 Biomechanical and physics-informed regularisation

A third group replaces a generic smoothness penalty with a statement about tissue mechanics or about the function space itself. Qin and colleagues learned the prior in place of writing it down: a variational autoencoder trained on biomechanically simulated deformations defines a manifold of plausible fields, and

tracking becomes a search over that manifold, validated on three public cine datasets [53]. WarpPINN states the physics explicitly. It represents the deformation with a neural network fitted per case under an intensity-similarity term plus a hyperelastic regulariser and a Jacobian penalty for near-incompressibility. Strain follows by automatic differentiation, and Fourier feature mappings allow sharp features in the strain field. It was tested on synthetic examples and a cine benchmark of 15 healthy volunteers [54]. WarpPINN-fibers adds a fibre-stretch penalty and reported improved landmark tracking and strain-curve prediction on a synthetic phantom and the same healthy benchmark [55]. Liu and colleagues combine the mechanical and temporal arguments in a space–time finite-element correlation framework with region-specific biomechanical regularisation and a data-driven temporal decomposition. Their evaluation is the most layered in this group: one synthetic dataset with ground-truth motion and strain, three public datasets with landmarks or masks and a clinical dataset with masks, against two feature-tracking and four deep-learning baselines [56]. Implicit neural representations of intramyocardial motion and strain have also been presented in the 2026 STACOM proceedings; no abstract accompanies the record retrieved, so that study is not characterised here [57].

Such results show that mechanics can be useful inductive structure. They do not imply that a constrained field is uniquely correct, because disease can alter wall thickness, loading, scar mechanics and fibre organisation, and a 2D slice omits through-plane motion. The limiting factor is shared across the group: where a strain reference exists at all, it exists in one simulation setting, and the in-vivo evaluation returns to landmarks and masks. A mechanical prior is a differently shaped constraint from a smoothness prior, neither weaker nor stronger in general, and whether it preserves a focal deficit depends on whether its constitutive assumptions hold in heterogeneous injured tissue. That dependence is an open question, and validation must test it directly by asking whether the constraint preserves the abnormality of interest, in addition to assessing plausibility.

3.6 Learning from cine, contours or motion-encoded CMR

The final group changes the supervision in place of the prior, and within it the methods differ chiefly in their target. DeepTag estimates motion on tagged images without labels, using bidirectional diffeomorphic registration between consecutive frames and a differentiable composition layer to recover the Lagrangian field, validated by landmark tracking on a clinical tagging dataset [58]. Networks trained on synthetic 3D tagged images from a biophysical left-ventricular model have since been proposed for 3D tagging analysis [59]; like DeepTag, they operate on the comparator acquisition and not on cine. DeepStrain is the most completely documented end-to-end cine workflow. It centres the ventricle, segments anatomy, estimates end-diastole-to-phase displacement and derives Green–Lagrange radial and circumferential strain, and the original study reported a CMAC landmark end-point error of 2.9 ± 1.5 mm together with scan–rescan repeatability in ten healthy volunteers and high agreement for volumetric measures [60]. A later single-centre comparison in 47 healthy participants and 533 patients found correlations with manual feature-tracking analysis of 0.83–0.85 for global radial and 0.88–0.91 for global circumferential strain [61]; in that comparison the comparator is another estimator. These results establish workflow and output properties. Independent dense material truth for the cine motion field is a separate matter, and the one prescribed-motion test of DeepStrain that addresses it is discussed in Section 4.1.

StrainNet is particularly relevant because DENSE phase-derived displacement was the supervised target, and its two evaluations must be read separately. Of 305 eligible individuals (161 patients, 99 healthy adults and 45 healthy children or adolescents), 243 individuals contributing 670 sections formed the training set and 62 individuals contributing 190 sections formed the independent test set; the 305 were not independent held-out cine tests. For pixelwise displacement end-point error (EPE), the model was

deliberately given contours derived from DENSE magnitude images, permitting one-to-one comparison with DENSE phase-derived displacement over the myocardium and time series; the independent-test EPE was 0.75 ± 0.35 mm. In the distinct routine-cine analysis, contours from bSSFP cine feature tracking were supplied to StrainNet and registered DENSE was the comparator. End-systolic global and AHA-segmental circumferential Green–Lagrange strain was analysed in 59 of the 62 test participants with basal, mid-ventricular and apical images: ICCs were 0.87 versus 0.72 globally and 0.75 versus 0.48 segmentally for StrainNet versus conventional feature tracking [62]. This is substantive positive evidence for bounded regional recovery. The bounds are these: the EPE is not a cine-contour result, "global" strain is not the pixelwise whole-myocardium EPE, and end-systolic values are not peak-strain timing. The inference therefore remains model-, direction-, population-, acquisition- and registration-specific, and it does not establish focal location or extent.

The same supervision route has since been extended. Lei and colleagues report a DENSE-supervised network for routine bSSFP cine that estimates large myocardial rotation in a separate time-series branch and radial motion by deformable registration in a second, motivated by the observation that earlier DENSE-supervised models captured twist and torsion poorly [63]. A pre-clinical study takes the idea to a different reference: MyoNet was trained to recover displacement from cine in a rat model of radiation-induced injury against strain derived by a sine-wave modelling method, reporting ICCs of 0.973 and 0.975 for circumferential and radial strain [64]. Both are abstract-level readings here, and the second, being a rodent study against an analysis-derived reference, bears on the design and not on human cine accuracy.

A further set of systems is supervised on a downstream label in place of motion. Cine-only infarct delineation used recurrent motion features and optical flow, supervised on manual delineation of delayed-enhancement images [65]. Deep-learning synthetic strain derived regional wall-motion assessment from cine and was evaluated against the consensus of four cardiothoracic radiologists in 53 patients and 846 segments, reaching an AUC of 0.90 at a 30% peak radial strain threshold [66]. Deep-learning aligned strain was developed to detect fibrotic myocardium in Duchenne muscular dystrophy with LGE as the reference [67], and Bello and colleagues supervised a motion representation directly on survival, reporting a Harrell C index of 0.75 against a human benchmark of 0.59 in 302 patients [68]. Each is a well-posed study, and in each the supervision target determines what the evidence can support, which in every case is something other than the accuracy of the motion field.

3.7 What the method record shows by year

Three patterns in Supplementary Table S3, plotted by year and family in Figure 3, define the rest of this review. First, the modal evidence across the methods reviewed, and across the 2018–2026 cardiac-specific designs in particular, is agreement between a propagated and a reference segmentation, supplemented by sparse landmark error; by the construction in Figure 2, that evidence cannot by itself certify material correspondence. Second, where a method is evaluated against a material-sensitive reference, the reference is almost always acquired in a mixed or healthy population, and the comparison is reported as agreement, without error under controlled conditions. Third, the pathology column is mostly "not reported": among the estimators reviewed, deliberate evaluation on a focal abnormality of known size and severity appears in one independent synthetic study (Section 4.1), and the evidence that does exist on pathological hearts tends to be mask agreement or a downstream label. These three patterns describe the published evaluation record; they say nothing about the methods' capability.

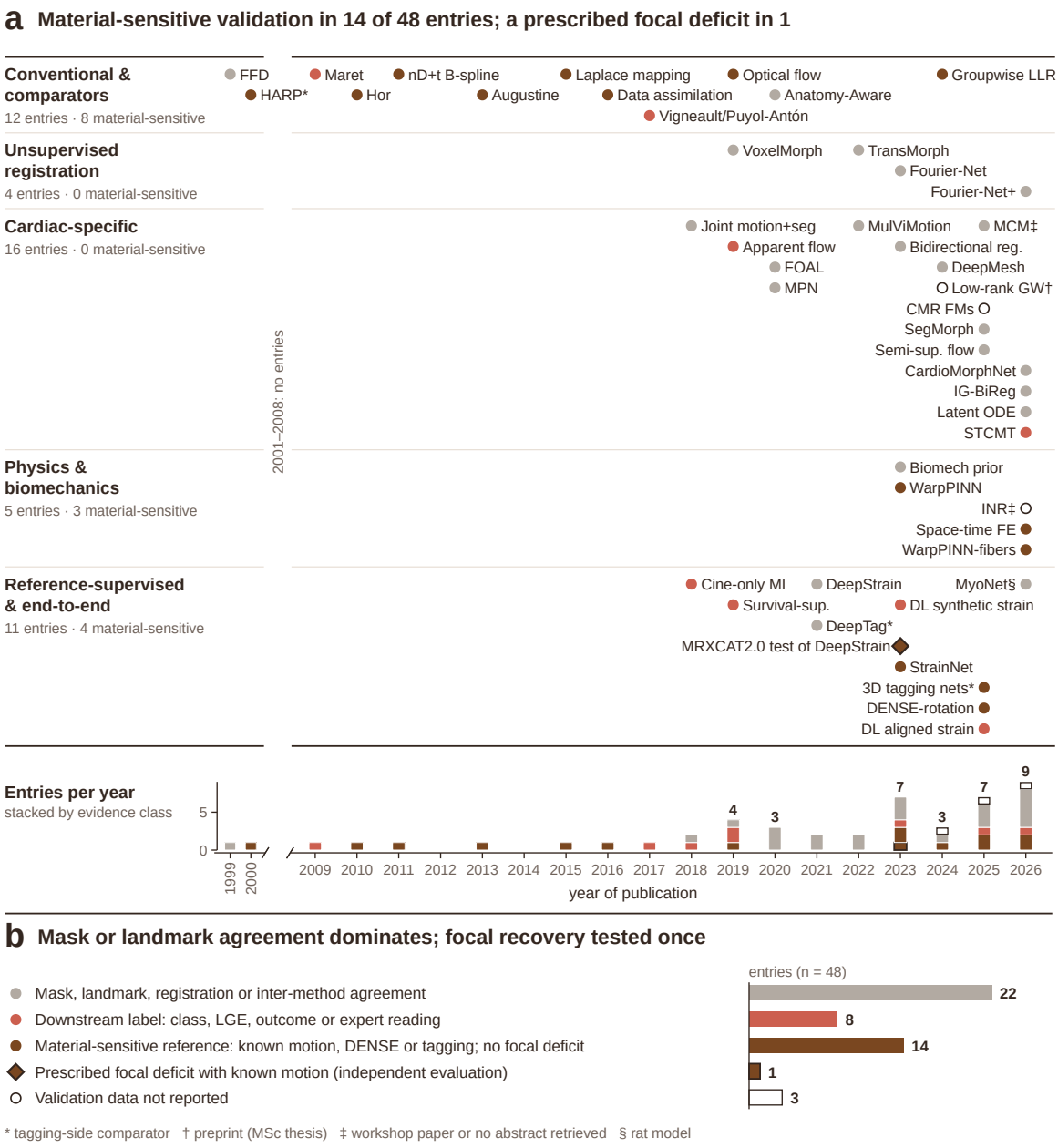


Figure 3. Cine-CMR motion estimators by year and by the strongest validation evidence each cited source reports (Supplementary Table S3). (a) Lanes group methods by family, with the number of entries and of material-sensitive entries under each lane name; marker fill encodes evidence type, and the stacked columns on the shared year axis count entries per year by evidence class. Most cardiac-specific designs from 2018 to 2026 report mask or landmark agreement; material-sensitive references (known motion, DENSE or tagging) appear in 14 of 48 entries, in healthy or mixed cohorts or in simulation; a prescribed focal deficit appears in one independent synthetic evaluation (the MRXCAT2.0 test of DeepStrain). (b) Evidence classes and their totals (22, 8, 14, 1 and 3 entries). Evidence classes were assigned from the Table S3 cells, which are abstract- or metadata-level readings. The 48 plotted entries comprise every Table S3 row except boundary feature tracking, a method family without a single year; the early clinical feature-tracking row is shown as its three studies, and the MRXCAT2.0 test of DeepStrain is shown separately from the DeepStrain row. Years 2001–2008 contain no entries and are compressed. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

4 What the validation literature establishes

No single study can answer every layer of the measurement chain, so the most informative synthesis groups evidence by validation target in place of algorithm chronology. Figure 5a summarises what each target can establish. The following subsections take the targets in turn, from the most controlled to the most clinical.

4.1 Known-motion tests establish capacity and technical error under controlled conditions

Numerical phantoms make exact correspondence available. A simple realistic phantom showed that spatial and temporal resolution can materially affect feature-tracking measurements [69]. MRXCAT2.0 couples statistical shape, biophysical simulation and tissue texture to produce healthy and pathological cine-like images paired with known functional ground truth [70]. Other in-silico heart models and DENSE-SIM extend controlled testing of strain pipelines and acquisition effects [71, 72]. The design that such resources make possible, a graded regional deficit measured against a material reference while the estimator is held fixed, has a precedent outside cine: Yeon and colleagues compared tagged circumferential shortening with implanted sonomicrometry crystals in a canine coronary-ligation model, with both methods separating ischaemic from remote myocardium [73]. That design is the one this review finds largely missing for cine estimators.

Within the cine literature retrieved, the most informative single prescribed-focal result is the DeepStrain evaluation reported in the MRXCAT2.0 paper, which is independent of the DeepStrain developers. Four cases were simulated (normal, dilated, hypertrophic and infarcted left ventricles, with ejection fractions of 51%, 34%, 41% and 49%), and the infarct was modelled as an elliptical free-wall scar. The infarcted ejection fraction stayed at 49% because remote myocardium compensated for the scar, which is precisely the configuration a global measure cannot see. The prescribed deficit was strongly attribute-dependent before any estimator was applied: ground-truth peak systolic strains were 0.95 (radial), −0.18 (longitudinal) and −0.18 (circumferential) in remote tissue against 0.30, −0.17 and 0.01 in the scar, so radial strain falls to about a third of its remote value, circumferential strain is near zero in the scar, and longitudinal strain is almost unchanged. Applying DeepStrain gave a mean Dice of 0.82 across phases, lowest for the infarcted case and particularly in the thin scar region, an average displacement error of 1.0 ± 0.9 mm, a mean circumferential strain error of 0.02 ± 0.04 averaged across the four cases, with larger errors in the thinner-walled dilated and infarcted cases, and a general underestimation of radial strain, -0.24 ± 0.21 averaged across cases with the lowest performance attributed to the infarcted case at -0.20 ± 0.21. All values are case-level averages over mid-ventricular short-axis slices; none is an error measured within the scar region itself [70]. Figure 4 plots the prescribed deficit and the reported errors side by side.

Two qualifications come from the source itself. The authors describe the infarcted case as the worst performer, yet the radial error they report for it (−0.20) is smaller in magnitude than the across-case average (−0.24) and the per-case standard deviations overlap, so the ordering rests on the full set of panels and should not be quoted as a quantitative ranking. The experiment also occupies a single point in the space this review is asking about, with one scar geometry, one estimator, mid-ventricular slices only and one simulated signal-to-noise ratio, and the record read does not identify which DeepStrain release or weights were used. What the experiment does establish is the empirical instance of the claim form developed in Section 5.3: on the same images, the same estimator kept the case-averaged circumferential error at 0.02 ± 0.04 while generally underestimating radial strain, most visibly where the wall was thin. Because the errors are averaged over whole slices, the experiment shows that the circumferential component survived better than the radial component, and it does not report how well the scar itself was localised or bounded. The result is attribute-specific. The single experiment separates none of the candidate explanations for it, which include training-set composition, wall thinning and partial volume in the scar, the generator's

image-formation model, and the estimator's own regularisation.

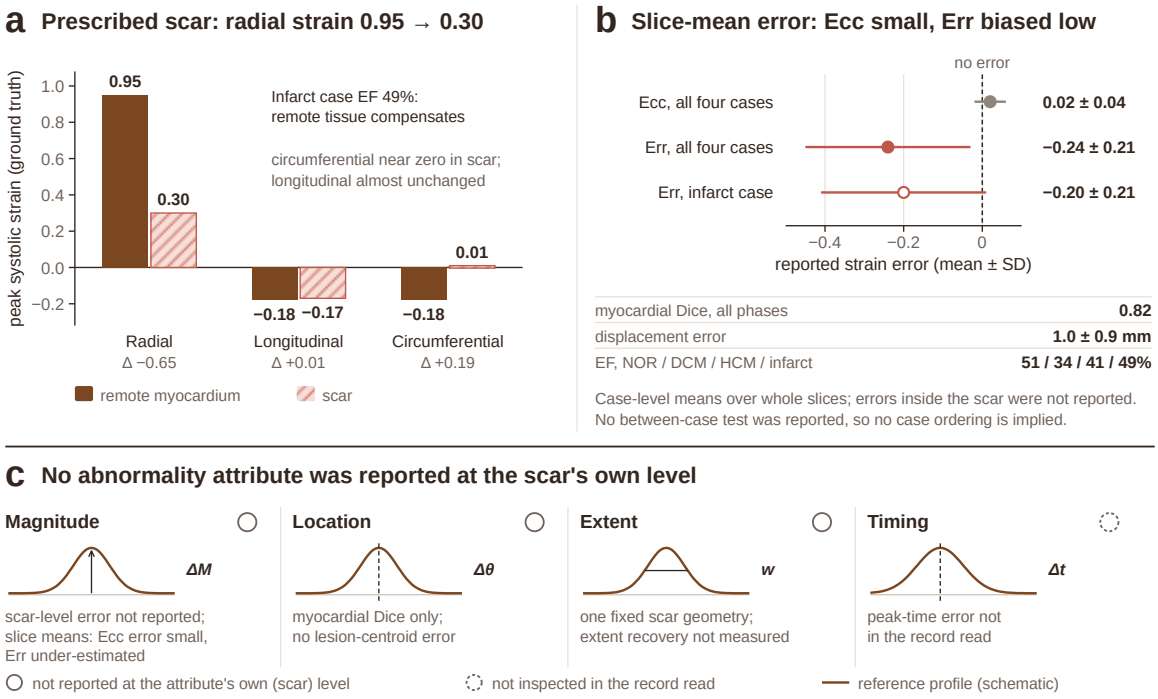


Figure 4. Attribute-specific recovery in the one independent prescribed-focal test. (a) Ground-truth peak systolic strain in remote and scar tissue of the MRXCAT2.0 infarct case: radial strain falls from 0.95 to 0.30 ($\Delta = -0.65$), circumferential strain is near zero in the scar ($\Delta = +0.19$), longitudinal strain is almost unchanged ($\Delta = +0.01$), and ejection fraction remains 49%. (b) DeepStrain error across the four generated cases: circumferential strain error 0.02 ± 0.04, while radial strain is generally underestimated (-0.24 ± 0.21; infarct case -0.20 ± 0.21); the same report gives myocardial Dice 0.82, displacement error 1.0 ± 0.9 mm and ejection fractions of 51, 34, 41 and 49% for the normal, dilated, hypertrophic and infarct cases. No between-case statistical test was reported and the per-case standard deviations overlap, so no case ordering is implied. Errors are case-level means over whole slices; errors within the scar were not reported, so a small slice-averaged error leaves the scar-level deficit untested. (c) Status of the four abnormality attributes in this experiment, each with a schematic reference profile marking what the attribute measures: none was reported at the level of the scar itself, and timing was not inspected in the record read. Values are as reported by the MRXCAT2.0 authors [70]; Δ is scar minus remote; one scar geometry, one estimator, mid-ventricular short-axis slices; no new analysis. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

Known-motion resources answer questions that an in-vivo comparison cannot: whether the representation can express a lesion, whether the optimiser recovers it at a chosen signal-to-noise ratio, and how sampling or regularisation changes magnitude and localisation. Along those lines, resolution has been varied against known motion [69], analysis regularisation has been varied against known motion [72], and abnormality has been represented in known-motion data [70, 74]. Within the searches recorded here, this review did not identify a study that varied abnormality size, severity or transmurality against known motion while holding the estimator and the measurement definition fixed; that absence is a property of the recorded searches, and it does not assert that no such study exists. The limitation of simulation is just as clear. A simulation is valid only over the anatomy, mechanics, appearance, artefacts and pathology it represents, so passing a phantom test establishes recovery in that regime and leaves in-vivo accuracy open. The appropriate response is to pair simulation with material-sensitive and clinical tests, which the next subsections take up.

4.2 Paired tagging or DENSE provides material-sensitive in-vivo evidence

Tagging introduces a prepared magnetisation pattern; DENSE encodes displacement in image phase. Both make myocardial motion more directly observable than routine cine, and neither is error-free. Tag fading, spatial filtering, phase unwrapping, signal-to-noise, through-plane motion, temporal fitting and registration to cine can all contribute reference error. A comparison should therefore report reference uncertainty and acquisition alignment, and it should avoid calling either technique literal ground truth.

The paired literature is mixed in a scientifically useful way. In three healthy volunteers and 41 patients with recent or chronic myocardial infarction, Gao and colleagues reported agreement in peak circumferential strain and similar regional patterns between cine registration and DENSE, while cine radial strain was overestimated [75]. A larger comparison of 77 patients with cardiomyopathy and ten controls found relatively small circumferential differences but wide radial differences across tagging, DENSE and three feature-tracking products [17]. Wehner and colleagues showed that part of the apparent cine–DENSE discrepancy can arise from comparing different strain definitions, while still finding that regional tracking information cannot be inferred from contours alone [16].

The product dependence seen by Cao and colleagues was examined directly at 3 T by Militaru and colleagues, who compared three feature-tracking products with tagging in 61 participants. Global longitudinal and circumferential agreement was high, regional and radial agreement varied more, and scar discrimination differed by vendor [21]. This pattern supports global and some regional use under specified products, and it warns against interchangeability and against extrapolating global performance to radial or highly local outputs; it does not justify a blanket rejection of feature tracking. Timing has been compared less often than magnitude. In 42 participants, Brandt and colleagues found moderate-to-good agreement for global and segmental time to peak strain between tagging and feature tracking, alongside significant biases in segmental peak strain [76]; this is a segment-level timing comparison, and focal peak-time error remains untested in it.

Recent work continues to show direction- and model-specific success. A 2026 study evaluated one- and two-slice DENSE models and a one-slice cine model using a deforming phantom and 40 healthy volunteers. Circumferential strain was recovered well in the phantom; in volunteers, the one-slice cine model was statistically equivalent to the two-slice DENSE model for endocardial circumferential strain within a prespecified margin of 3.5% (two one-sided tests) and differed for radial, longitudinal and epicardial circumferential estimates [77]. The result identifies which component, layer and setting were supported and where the slice model was insufficient; it supports neither “cine works” nor “cine fails” as a general verdict. Selected studies are compared directly in Table 2.

Table 2. Empirical validation studies relevant to regional fidelity. The analysis unit and support boundary are retained because participant, slice, segment and pixel results are not interchangeable.

Study; development/test denominator	Population/input; statistical unit	Spatial/temporal estimand	Comparator	Result, including uncertainty where reported	Supported scope
Wehner et al.[16]; 88 subjects, 186 paired slices	Paired cine/DENSE at two centres; subject/slice	Whole-slice base/mid/apex Ecc; longitudinal strain, torsion and dyssynchrony	DENSE; contour-only calculation	Mid-slice Ecc bias −0.4% and coefficient of variation 11%; longitudinal strain bias −0.2%, coefficient of variation 19%; dyssynchrony bias 26 ms, coefficient of variation 76%	Selected whole-slice quantities, not focal localisation or extent
Cao et al.[17]; 87 subjects	77 cardiomyopathy, 10 controls; subject/mid-slice segment	Peak Ecc, Err and Ell from DENSE and three FT products	Tagging	DENSE Ecc agreement with tagging ICC 0.778; technique/product differences were much wider for Err than Ecc/Ell	Peak strain in one mid slice; no dense location/extent error

Study; development/test denominator	Population/input; statistical unit	Spatial/temporal estimand	Comparator	Result, including uncertainty where reported	Supported scope
Militaru et al.[21]; 61 subjects	18 healthy, 18 chronic MI, 15 DCM, 10 LV hypertrophy; subject/segment	Global and 16-segment transmural peak systolic LS/CS/RS	Tagging; LGE in MI	Versus tagging, global ICC 0.92–0.94 for LS and 0.88–0.91 for CS; global RS ICC across vendors 0.10–0.81; regional differences were larger	Strong global LS/CS under tested products; weaker regional/radial interchangeability
Kihlberg et al.[78]; 116 patients	Suspected CAD; subject/segment	Global/segmental circumferential/radial strain and torsion	LGE > 50% transmurality	34 patients had qualifying scar; segment AUC 0.87 DENSE, 0.83 tagging, 0.66 FT; sensitivity at 80% specificity 82%, 71%, 35%	Scar discrimination at the stated threshold, not material-motion error
StrainNet [62]; 243/62 subjects (670/190 sections)	Patients; healthy adults and children; pixel, subject and AHA segment	DENSE-contour displacement over time; separately cine-contour end-systolic Green–Lagrange Ecc	DENSE phase displacement; conventional FT	Test EPE 0.75 ± 0.35 mm with DENSE-magnitude contours; in 59 cine tests, ICC 0.87 vs 0.72 global and 0.75 vs 0.48 segmental	Bounded Ecc support; EPE is not ordinary-cine displacement error
Strain-8 [79]; 20 volunteers	Healthy same-day scan–rescan; subject/segment	Global and segmental peak strain across eight protocols	Repeat acquisition	Global coefficients of variation ranged from excellent to fair ($\leq$ 20%); segmental values were frequently poor (> 20%), especially longitudinal/radial	Same-day precision in healthy volunteers, not accuracy
MRXCAT 2.0 with DeepStrain [60, 70]	Generated normal, dilated, hypertrophic and infarcted cases (EF 51/34/41/49%); mid-ventricular short-axis slices	Known simulated peak radial, circumferential and longitudinal strain; pixel/mesh	Biophysical prescription with an elliptical free-wall scar	Dice 0.82; displacement error 1.0 ± 0.9 mm; mean Ecc error 0.02 ± 0.04 and mean Err error −0.24 ± 0.21 averaged across cases and slices (−0.20 ± 0.21 for the infarct case)	Component-specific capacity/error within the generated domain; slice-averaged, with no scar-region error reported; one scar geometry and one estimator; release not identified
Jahromi et al.[77]; phantom and 40 volunteers	Healthy, mid-ventricular; subject/layer/slice	Slice-wise Ecc, Ell and Err over normalised cycle	Deforming phantom; one-/two-slice DENSE models	Cine agreed reasonably for endocardial Ecc; radial/longitudinal magnitude was higher and epicardial Ecc magnitude lower than DENSE models	Component- and layer-specific equivalence, not pathological dense motion

The studies in Table 2 do not all inspect the four attributes in the motivating question, so Table 3 records the direct endpoint of each, keeping a segmental analysis distinct from a localisation or extent validation.

Table 3. Direct inspection of abnormality magnitude, location, extent and timing in selected paired or regional studies. An entry of "not inspected" records that the study did not evaluate the attribute; it says nothing about whether the attribute is present or can be evaluated.

Study	Magnitude	Location	Extent	Timing
Gao et al.[75]	Peak regional circumferential/radial strain; circumferential agreement, radial overestimation	Similar regional patterns reported at abstract depth	Lesion-boundary or width error not inspected	Peak values; timing error not inspected
Wehner et al.[16]	Base/mid/apex whole-slice Ecc, longitudinal strain and torsion; only mid-slice Ecc met the study's accuracy criterion	Slice level, not focal-lesion localisation	Focal spatial extent not inspected	Dyssynchrony was direct and showed poor cine–DENSE agreement

Study	Magnitude	Location	Extent	Timing
Militaru et al.[21]	Global and segmental peak strain; direction/product-dependent agreement	Segment-level scar discrimination was tested and varied by product	No lesion-boundary or lesion-size error	Peak endpoints; timing error not a prespecified comparison
Kihlberg et al.[78]	Global/segmental strain and torsion associated with LGE category	Segment-level discrimination of high-transmurality LGE was direct	A > 50% transmurality category is not a boundary/area error	Timing error not inspected
StrainNet [62]	End-systolic global and segmental Ecc showed positive DENSE agreement	AHA segment averages; no focal localisation-error endpoint	Focal extent not inspected	Fixed end-systolic comparison; peak-time error not inspected
DeepStrain on MRXCAT2.0 [70]	Prescribed scar: case-averaged Ecc error 0.02 ± 0.04 versus Err error -0.24 ± 0.21; the only prescribed focal deficit in the studies reviewed, reported as slice averages with no scar-region error	Lowest Dice reported in the thin scar region; no centroid or boundary-distance endpoint	Scar extent fixed by design; extent error not reported	Peak values; timing error not inspected
Jahromi et al.[77]	Component-, layer- and slice-specific strain in phantom/healthy volunteers	No pathological lesion-localisation task	Focal extent not inspected	Focal timing error not a reported endpoint

4.3 Repeatability establishes precision, and leaves accuracy open

Scan–rescan studies are essential for longitudinal use, because they show how much a result may change when the underlying subject is assumed stable. The Strain-8 study scanned 20 healthy volunteers twice on the same day using eight protocols: cine, tagging and DENSE at 1.5 and 3 T, plus 2D and 3D echocardiography. Global repeatability ranged from fair to excellent depending on direction, protocol and field strength, while segmental results were more variable [79]. A multicentre travelling-volunteer study found the same kind of variation across scanners and post-processing [80]. Two cautions follow from such studies. A repeatable biased method can still be inaccurate, and a method with good global repeatability can have weaker segmental precision. Both absolute error and variance should therefore be reported at the claimed scale.

4.4 Tissue references test biological correspondence, and leave motion truth open

LGE is an independently measured tissue characteristic with strong relevance to infarction, and it is not a displacement measurement. If regional strain differs between scarred and non-scarred segments, the result supports tissue association and possibly diagnostic discrimination. It leaves the pointwise displacement error unknown, because scar transmurality and deformation are related without being identical, and remote loading, tethering and compensation can alter strain.

Kihlberg and colleagues provide a clear example. In 116 patients with a high likelihood of coronary artery disease, 34 had more than 50% transmural LGE in at least one segment. For discriminating those segments, reported AUCs were 0.87 for DENSE, 0.83 for tagging and 0.66 for feature tracking; at 80% specificity, sensitivities were 82%, 71% and 35%, respectively [78]. This is clinically relevant comparative evidence that the techniques captured scar-associated dysfunction differently in that cohort. Read as a direct ranking of material-motion accuracy, it would say more than it measured.

Other studies have detected regional feature-tracking abnormalities within infarcted tissue or related regional strain to recovery and outcome [3, 81]. Such findings strengthen biological and clinical qualification. They also illustrate why motion accuracy and clinical usefulness should both be studied: a regionally smoothed estimate might retain prognostic signal, whereas a physically accurate field may fail to improve a decision if the endpoint or treatment pathway is poorly specified.

4.5 Outcome association is downstream evidence

Global or regional strain has been associated with adverse events across disease cohorts [5, 6]. Such associations establish prognostic value under the model, covariates and follow-up that were studied. Whether changing management on the basis of the measurement improves outcomes is a further question, and answering it requires a decision context, a comparator, a threshold, the consequences of error and preferably an impact evaluation. The distinction matters most for a regional map, where multiple testing, correlated segments and post hoc threshold selection can produce apparently focal findings.

5 Reconciling apparently contradictory findings

The literature looks contradictory when all endpoints are treated as interchangeable. Once differences in input, estimand and target are made explicit, the same literature becomes more coherent, and the following four subsections work through the main sources of apparent contradiction.

5.1 Global agreement and regional disagreement can coexist

Spatial averaging suppresses zero-mean local errors. If one segment is overestimated and another underestimated, a global mean may remain accurate. Boundary-constrained deformation can also reproduce total shortening while distributing it differently around the wall. For both reasons, strong global correlation and small global bias do not mathematically imply correct location or extent of a regional abnormality. Regional validation must therefore evaluate regional errors directly and account for correlated repeated segments within participants.

The reverse is also possible. A method may preserve the location of a focal reduction while showing a scale bias in absolute strain. Whether that output is useful depends on the intended task: localisation, longitudinal change, diagnostic threshold or mechanistic interpretation. A single correlation coefficient conceals these possibilities. At minimum, studies should show bias and limits of agreement, spatial localisation error, extent error, timing error and subject-level distributions.

5.2 Radial strain is a special stress test

Radial strain is repeatedly less consistent than circumferential or longitudinal strain [17, 21]. Three features of the measurement explain much of this. The ventricular wall spans relatively few voxels radially, trabeculation complicates the endocardial boundary, and a small contour displacement can produce a large relative change in thickness. Products may additionally calculate radial output from different entities. These factors make the definition and uncertainty of radial strain unusually important, without making radial function unknowable. Claims should therefore specify whether the result is tensor radial strain, wall thickening, a layer-to-layer distance change or another surrogate.

5.3 Regularisation can help and can bias

Smoothness, temporal bases, incompressibility and learned priors stabilise an ill-conditioned estimate. They can reduce noise and prevent folding, and they may also spread or attenuate a focal abnormality if the imposed scale is too coarse. Which of these effects dominates is an empirical matter: a smooth representation with sufficient basis functions can represent a focal defect, and an unconstrained dense field can still miss it because image cues are weak or optimisation fails. Three questions should therefore be separated:

1. **Representation capacity:** can any parameter setting express the target motion at the required scale?
2. **Estimation:** given images that contain sufficient cues, does the optimiser or trained model recover those parameters?

3. **Input sufficiency:** do routine cine images contain enough information, under the tested acquisition and pathology, to distinguish the relevant motions?

Each question needs its own test. A representation-capacity test should not be replaced by a registration score, and failure on clinical cine should not automatically be attributed to the basis. In the other direction, success on noiseless synthetic images does not establish that the clinical input contains discriminating information. Section 3.1 gives the experiment that separates the three, and the DeepStrain evaluation on MRXCAT2.0 in Section 4.1 shows why it is needed: the same estimator on the same images preserved one strain component better than another at the slice-average level. Nothing in that single experiment says which of the three was responsible.

5.4 Uncertainty and quality control answer a different question

An uncertainty map can describe posterior dispersion, ensemble variability or sensitivity to perturbation. It becomes a clinically interpretable error estimate only after calibration against relevant reference errors. Quality control stands in the same relation to accuracy: it can detect image artefact, contour failure or field folding, and it cannot guarantee that an accepted case has accurate regional strain. Methods should therefore report the observable signal used by a quality rule, its coverage, and error among accepted and rejected cases. A selective or abstaining workflow is appropriate only when such a rule is actually implemented, and it should not be imposed as a rhetorical requirement on every estimator. The consolidated mapping from validation target to supported claim is provided in Supplementary Note 2.

6 A validation programme for regional or focal claims

A persuasive evaluation should narrow from controlled capacity to clinical relevance while keeping the measurement contract fixed. Its objective is to match the strength of the claim to the evidence provided, which does not require every possible experiment from every paper. Figure 5b shows the resulting evaluation flow, and the subsections below describe its stages.

6.1 Pre-specify the measurement contract

Before comparing methods, investigators should state the cine protocol, views, temporal and spatial resolution, reconstruction, image-quality exclusions, abnormality type and scale, estimator version, user interaction, strain definition, reference alignment, target attributes and tolerances. Case splits should occur at participant and, where relevant, centre level. Repeated segments should not be analysed as independent observations. Hyperparameters, model selection and any quality threshold should be chosen without using the final test set. Competing methods should be tuned on the same data with comparable effort, because a retrained baseline and a published score answer different questions; and releasing the transformation direction, differentiation code and strain definition makes a disagreement diagnosable where a single aggregate score does not.

The primary endpoint should be an error in the claimed attribute, in preference to the easiest available surrogate. If the claim is detection of a focal region, suitable endpoints include centre-of-abnormality distance, overlap or surface distance of the abnormal region, with sensitivity across lesion sizes. If the claim is quantitative severity, bias and limits of agreement for regional magnitude are required. If timing matters, peak-time or curve-alignment error should be assessed in addition to peak amplitude.

6.2 Stage 1: challenge representation and sampling with known motion

Known-motion experiments should vary one factor at a time and then test interactions: abnormality width, transmural extent, amplitude, onset/peak timing, position, wall thickness, through-plane motion, temporal resolution, spatial resolution, contrast, noise and artefact. The true field should be projected

into the candidate representation to measure approximation error independently of the estimator; this is the representation-capacity test of Section 3.1. The estimator is then run on rendered images to quantify additional optimisation or learning error.

An equal-mean counterexample is particularly useful at this stage (Figure 5c). Two fields can have the same global strain while one contains a focal abnormality. Recovery of their mean is then no evidence of success; the method must distinguish their location and extent within pre-specified tolerances. Such tests show whether apparent global success masks spatial redistribution.

6.3 Stage 2: use paired material-sensitive in-vivo acquisitions

The next stage should pair routine cine with tagging or DENSE in the same session and align acquisitions carefully. Reference processing should be blinded to cine outputs where feasible. Registration between acquisitions, slice mismatch and temporal alignment should be quantified. Analyses should report results separately for healthy myocardium and relevant disease patterns, including scar location, wall thickness and image quality. External centres and scanners test transfer only if participants and acquisition conditions are genuinely independent of training and model selection.

No material-sensitive reference is perfect, so this stage also needs a sensitivity analysis. Varying reference smoothing, registration and segmentation estimates how much apparent cine error is attributable to the comparator. Where tagging and DENSE disagree, the disagreement should be reported; choosing the more favourable reference would hide it.

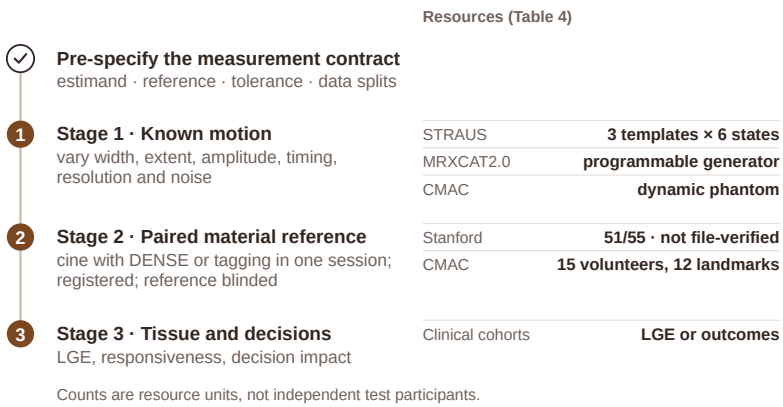
6.4 Stage 3: connect motion fidelity to tissue and decisions

Tissue characterisation, expert wall-motion scores and outcomes become valuable after the technical claim is clear. LGE can test whether a regional measurement captures scar-associated dysfunction; perfusion or mapping can address other substrates. Longitudinal studies can test responsiveness and minimal detectable change. Decision studies should specify what action changes, how false-positive and false-negative regional calls matter, and whether the regional measurement adds information beyond volumes, LGE and clinical covariates.

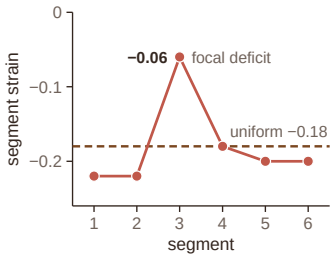
a Each validation target supports one claim, not the next

Validation target	✓ Strongest claim it supports	✕ Does not establish on its own
Known motion phantom, STRAUS, MRXCAT2.0	representation capacity and technical error in the simulated regime	in-vivo accuracy
Paired DENSE or tagging same session, registered	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan–rescan same subject, re-imaged	precision; accuracy needs a reference	bias
LGE or tissue characterisation scar or fibrosis label	biological association; motion error stays unmeasured	pointwise displacement error
Outcome events, prognosis	prognostic association; utility needs a decision study	decision impact

b Pre-specify, then escalate the reference stage by stage



c Same mean, focal deficit



segments 1–3, focal	**−0.22 · −0.22 · −0.06**
segments 4–6, focal	**−0.18 · −0.20 · −0.20**
global mean, both fields	**−0.18**
range, focal / uniform	**0.16 / 0**

Recovering the mean proves nothing about the deficit.

Figure 5. What each validation target can establish, and a staged programme for regional or focal claims. (a) Five targets, the strongest claim each supports, and what each leaves open on its own. (b) Pre-specification followed by three stages, with the resources reviewed in Table 4 placed at the stage they can serve; counts denote resource units and are not counts of independent test participants. (c) Two fields with an equal global strain of −0.18 and different regional content (segment values −0.22, −0.22, −0.06, −0.18, −0.20 and −0.20 against a uniform −0.18; range 0.16 against 0): recovering the mean is no evidence of regional recovery. Conceptual synthesis; no study data. Drawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic); no generative image model was used. Author verification is required before submission.

6.5 Validation resources answer different parts of the question

No available resource simultaneously supplies routine cine, error-free dense in-vivo motion, broad pathology and unrestricted external testing, so the value of the existing resources lies in combination. STRAUS supplies controlled multimodal images and framewise reference meshes; the Stanford repository supplies human acquisitions with closely matched cine, DENSE and tagging locations; CMAC supplies a historical cross-method benchmark with sparse material-sensitive landmarks; and MRXCAT generations supply programmable motion and acquisition conditions. For each, provenance and the unit of independence matter as much as file availability (Table 4).

STRAUS contains 18 *virtual patients*. These arise from three volunteer-derived multimodal templates, each driven through one healthy and five pathological electromechanical states (left bundle branch block, left-anterior-descending, left-circumflex and two right-coronary-artery ischaemia patterns), so the 18 cases share three anatomies. Each virtual case provides 3D cine MR, three tagged-MR orientations, ultrasound and framewise meshes; the paper used engineering strain for its regional curves [74, 82]. The meshes therefore support known-correspondence displacement and strain calculations when coordinate system, reference frame, mesh identity and temporal sampling are matched. What they cannot supply is the full

distribution of human anatomy, disease, image artefact or model error.

The Stanford Cardiac Performance MRI resource has a complementary strength, human acquisitions. Its laboratory page describes 51 healthy volunteers scanned at 3 T from 2023 to 2025, with whole-LV bSSFP cine and DENSE, tagging and mapping at specified corresponding locations [83]. DataCite metadata updated on 27 July 2026 describes 55 participants (the same 51 healthy volunteers plus four people with aortic stenosis) under DOI 10.25740/xt223hz2438. The two counts are versioned descriptions of one repository. Neither total proves that every sequence is complete and usable for every participant, that all cine–DENSE slices are registered analysis pairs, or that a derived paper has an independent test cohort. The DENSE magnitude and phase DICOMs also require reconstruction, phase unwrapping, displacement modelling, segmentation, spatial/temporal alignment and an explicit strain definition before they become a reference, so they are material-sensitive observations and not ready-made labels. No subject files were downloaded or processed for this review.

CMAC paired cine SSFP, 3D tagged MRI and 3D ultrasound from a dynamic phantom and 15 healthy volunteers. Its human reference comprised 12 landmarks (four walls at three LV levels) manually tracked by two observers in 3D tagging, with a reported median inter-observer variability of 0.84 mm [84, 85]. That reference supports correspondence error at those landmarks and times, and it says nothing about dense strain accuracy between them. The MRXCAT family differs again. Original MRXCAT uses the XCAT phantom to generate cine or perfusion CMR with selectable cardiac/respiratory motion, sampling, coils and noise, and the 2014 demonstration was primarily an acquisition/reconstruction simulator [86]. MRXCAT2.0 adds statistical LV shape, biophysical healthy/infarct/dilated/hypertrophic function and generated tissue texture, exposing known simulated function for method testing [70, 87]. Both are generators with exemplar outputs, so neither has a fixed number of independent participants.

Large multi-centre resources such as CMRxRecon2025 remain useful for reconstruction robustness and disease representation. Its official description covers multiple centres, scanners, disease groups and CMR contrasts, and it does not establish tagging/DENSE motion labels or a fully paired cine–LGE cohort [88]. The same limit applies to segmentation collections and the Data Science Bowl: containing cine images does not make a collection a regional-motion reference. Detailed access, pairing, scale and uncertainty fields are provided in Supplementary Table S2 and the editable resource table.

Table 4. Complementary validation resources for cine-derived motion and strain. Counts denote the resource unit stated; they are not counts of independent test participants or complete paired analyses.

Resource	Nature and scale	Available pairing/reference	What it can evaluate	Access and principal boundary
STRAUS [74, 82]	Synthetic; 18 virtual cases from 3 templates × 6 functional states	3D cine, three tagged MR sequences and US with framewise VTK meshes	Dense correspondence, engineering strain, sampling and multimodal consistency in the simulated domain	Public, no login reported; only three source anatomies and imposed electromechanics/appearance
Stanford Cardiac Performance MRI [83]	Human; page: 51 healthy; DOI metadata: 55 total including 4 aortic stenosis	Whole-LV bSSFP cine; DENSE at matched locations; mid-LV tagging and other contrasts	In-vivo cross-sequence displacement/strain after explicit processing and alignment	Public repository, ODbL 1.0; completeness and usable pair count not yet file-verified
CMAC [84, 85]	Dynamic phantom and 15 healthy volunteers	Cine SSFP, 3D tagging and 3DUS; 12 manually tracked tagging landmarks	Sparse point correspondence and cross-modality benchmarking	Open access with attribution; sparse healthy reference, median inter-observer variability 0.84 mm
MRXCAT [86]	XCAT-based numerical generator, not a fixed human cohort	Prescribed anatomy/motion with cine/perfusion signal, coils, noise and sampling	Acquisition and reconstruction tests with known simulated motion	Code by non-commercial request; realism bounded by XCAT and signal assumptions

Resource	Nature and scale	Available pairing/reference	What it can evaluate	Access and principal boundary
MRXCAT2.0[70, 87]	Generator for healthy, infarcted, dilated and hypertrophic LV cases	Shape model, biophysical simulation, texture and known functional values	Capacity and estimation error across generated anatomy and patho-physiology	Public repository; generated instances are not independent clinical participants
CMRxRecon 2025 [88]	Multicentre, multi-vendor reconstruction resource with disease labels	Multiple CMR contrasts and reconstruction/QC material	Reconstruction robustness and domain shift	Challenge access terms; no declared dense motion truth or universal cine–LGE pairing

6.6 Reconstruction evidence is relevant and endpoint-specific

Reconstruction changes the images supplied to a downstream estimator, and primary studies show both agreement and residual bias. REGAIN was trained on 1616 patients and tested prospectively in 181 participants; its paired breath-hold comparison used 94 patients and 36 healthy participants, while global radial, circumferential and longitudinal feature-tracking strain was available in 49 participants. Mean global-strain differences versus a separately acquired GRAPPA cine were near zero, while the 95% limits of agreement remained −7% to 6% for radial, −2% to 3% for circumferential and −3% to 3% for longitudinal strain [89]. CineVN compared prospectively accelerated acquisitions with conventional GRAPPA in 18 healthy volunteers. On individual mid-ventricular slices, CineVN had the smallest bias and narrowest limits among the tested accelerated reconstructions, yet all accelerated protocols significantly underestimated peak radial strain and overestimated peak circumferential strain; its 46-patient demonstration lacked a conventional reference and reported slice exclusions after segmentation failure [90]. These are measured reconstruction-plus-analysis results. Whether deep learning removes or preserves a focal abnormality is a separate question that they do not address.

Other designs bound the inference differently. In ten healthy volunteers, model-based deep-learning reconstruction improved the precision of global circumferential strain relative to compressed sensing at reduction factors 6 and 8 when full sampling was used as comparator, while circumferential bias and substantial radial underestimation remained and no regional lesion was tested [91]. In 19 patients with 51 compressed-sensing acquisitions, retrospective temporal interpolation changed peak longitudinal and circumferential strain while ventricular volumes and ejection fraction stayed similar [92]; stable volumes do not guarantee stable strain. DENT was developed from 5840 patients across nine centres and compared interpolated with acquired cine phases, including prospective image-quality comparisons in 49 patients and 12 healthy participants. It reported image-domain interpolation error and reader assessment; myocardial strain and focal-abnormality recovery were outside its evaluation, and even its deformation vectors were not validated as motion estimates [93]. Reconstruction and motion estimation should therefore be evaluated jointly when the claim is regional, and any proposed mechanism (temporal smoothing, sharpening, alias suppression or hallucination) remains a hypothesis until tested against an attribute-appropriate reference.

7 Reporting and interpretation in practice

The consolidated reporting template is provided in Supplementary Note 5. Its essential logic is to identify the acquisition and reconstruction, estimator/version and interaction, complete strain estimand, statistical unit, validation target, failure policy and tested domain in the same record. For a focal claim, magnitude, location, extent and timing should be reported when each is relevant to the intended use, and an attribute that was not inspected should keep that label.

The same record serves two audiences. For clinical readers, a regional strain map should be read together with the cine images, contours, tissue characterisation and global function. For technical readers, improving Dice, smoothness or Jacobian regularity is worthwhile, and the improvement should be

connected to the regional quantity that motivates the method. For both, a negative result should identify which link failed: insufficient input cues, inadequate representation, failed estimation, mismatched strain definition, reference uncertainty or lack of clinical association.

Software status needs the same discipline. Software clearance and commercial availability do not substitute for analytical validation of a specific regional output, and the existence of reference ranges does not make values interchangeable across products. Serial assessment is safest when acquisition, software, version and analysis conventions remain stable, and any change should be bridged empirically.

8 Limitations of the evidence and of this review

The evidence base is heterogeneous. Many studies are single-centre or use modest samples, healthy volunteers or a restricted disease spectrum. Paired DENSE/tagging studies face acquisition burden and alignment error. Commercial algorithms are not always described sufficiently to reproduce the exact mapping. Studies often report global or segmental peaks without the full field, and regional analyses may treat segments as independent. Publication bias may favour successful methods and well-behaved images.

The review itself has matching limits. It used targeted searches, did not perform duplicate screening or risk-of-bias scoring, and did not pool effect sizes. Some recent 2026 method papers were available only at abstract or publisher-summary depth; they are used for publication status, design and high-level validation claims, without any unreported numerical conclusion. Product and regulatory details were kept outside the central narrative because versions and indications change rapidly. The accompanying ledger records reading depth and unresolved items so that absence of inspection is not mistaken for absence of evidence.

The proposed attribute-based framework is an organising approach; it is neither an established regulatory standard nor a new validated biomarker property. Tolerances must be justified for each intended use. No review can determine from published aggregate results whether a particular patient's regional map is correct.

9 Conclusions

Current evidence most directly supports cine-derived circumferential strain at whole-slice, global and AHA-segment scales when acquisition, software, myocardial layer, reference state and temporal endpoint are matched. Within those bounds, several paired studies found useful agreement with tagging or DENSE, and StrainNet improved end-systolic global and segmental Ecc agreement over the tested conventional feature-tracking pipeline. Scan–rescan and scar/outcome studies add evidence about precision and biological or prognostic relevance. Each of these study types supports use for its stated endpoint, and none converts that endpoint into dense material-motion accuracy.

Support weakens as the requested output becomes more local or changes definition. Radial strain is consistently more product- and method-dependent. Basal/apical, endocardial/transmural, two-/three-dimensional, engineering/Green–Lagrange and end-systolic/peak values cannot be exchanged without qualification. The one independent prescribed-focal test retrieved, DeepStrain on MRXCAT2.0, found a component-specific result (case-averaged circumferential error 0.02 ± 0.04, radial strain generally underestimated) from one scar geometry, one estimator and slice-averaged endpoints that do not quantify recovery of the scar region itself. In the studies reviewed, direct tests of focal abnormality centroid, lesion-boundary or width error and transmural extent recovery were uncommon or not reported. Timing has been compared mainly at global and segmental level (time to peak between tagging, feature tracking and speckle tracking in 42 participants [76]), and focal peak-time error against a material reference remains untested. These are statements about the inspected evidence, and they do not prove that no such work exists. The observed differences in sampling, software and strain definition explain some discordance.

Explanations based on information insufficiency, smoothing or low-dimensional bias remain hypotheses until a study varies those factors or measures representation capacity directly.

A defensible regional claim should therefore name the acquisition, input, estimator, population, spatial unit, myocardial layer, reference state, time point, comparator and tolerance. Known-motion resources such as STRAUS and MRXCAT2.0 can test capacity and estimator error; CMAC contributes sparse cross-modality correspondence; and paired human cine–DENSE/tagging resources such as Stanford can test transfer once explicit processing and alignment are done. Combining these complementary tests with participant-aware statistics and downstream clinical evaluation can identify where routine cine CMR preserves abnormality magnitude, location, extent and timing, and where the appropriate conclusion remains unresolved.

Declarations

Ethics approval and consent to participate

Not applicable. This review did not collect new participant data.

Consent for publication

Not applicable to the review content. The corresponding author must confirm that all author and institutional details are approved before submission.

Availability of data and materials

No new participant dataset was created. The review is based on published sources. The accompanying author-review package contains the search record, evidence ledger, figure sources and reproducible manuscript source; these materials document the narrative search and verification described in Section 1.1 and the Supplementary Methods.

Competing interests

Author confirmation required before submission. This draft does not certify the absence or presence of competing interests.

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Author contributions

The current title page lists Ziying Huo as the sole author. Authorship, order and contribution wording must be confirmed against the journal's criteria before submission. No contribution by a supervisor or other person is inferred from background materials.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with query execution, deduplication and screening suggestions for the structured search described in Section 1.1, bibliographic and source checks, evidence structuring, manuscript and table revision, language editing, LaTeX preparation and deterministic quality assurance. Claude Code (Anthropic) was subsequently used to merge the v8.1 and v9 drafts, restore the method-lineage and prescribed-focal-test content and their previously verified references, add cross-references and Supplementary Table S3, and run LaTeX, BibTeX and static consistency checks. No raw cardiac

images were processed and no new experimental result was generated. All figures and the graphical abstract in this revision were redrawn programmatically (Python-generated SVG rendered to PDF with headless Chromium) with AI assistance (Claude Code, Anthropic) from the manuscript text and tables; no generative image model was used, and no patient image, published figure or experimental result is reproduced. Editable sources, build scripts and render audits accompany the figures. The corresponding author must review and verify all content, citations, figures and declarations, confirm the journal's live policy interpretation and accept responsibility before submission; that review is not asserted here.

Figure provenance and permissions

All main, supplementary and graphical-abstract figures are original conceptual, analytic or numerical schematics drawn programmatically for this revision from the manuscript text, Supplementary Table S3 and the values reported in the cited MRXCAT2.0 paper. No patient image, measured strain map, algorithm output or third-party artwork appears in any figure. The editable SVG sources, the data files behind Figures 3 and 4 and the render audits are included in the author package. The corresponding author must inspect every figure, confirm its scientific content, and confirm the journal's current policy on AI-assisted programmatic artwork before submission.

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